

Summer of Innovation

Coding Club, IIT Dharwad

Public Test Cases

Intelligent Route Planning & Adaptive Optimization System

Part A: Sightseeing Route Optimization — Part B: Dynamic Ride Sharing with Flexible Routing

These test cases are provided to participants for verifying their implementations.

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Part A: Sightseeing Route Optimization

Background. The objective is to design an intelligent route planning system that determines an optimal sightseeing path from a given start to a destination, selecting a subset of intermediate locations to visit.

Satisfaction Decay Model: $S_{\text{eff}} = S \times e^{-k \cdot d}$, where S is the original score, d is the cumulative distance before reaching the location, and $k = 0.1$.

Category Penalty: If the number of locations visited from the same category exceeds threshold n , multiply that location's effective score by 0.9.

Objective: Maximise total effective satisfaction subject to the distance budget and category diversity constraints.

1.1 Public Test Case 1 *Easy*

Parameters

Start	Start
End	End
Distance Budget	20 km
Category Threshold (n)	2

Locations

#	Name	Score	Category
0	Start	—	START
1	Museum	8	Historical
2	Park	6	Nature
3	End	—	END

Distance Matrix (km)

	Start	Museum	Park	End
Start	0	3	5	10
Museum	3	0	4	7
Park	5	4	0	5
End	10	7	5	0

Task. Find the route from Start to End that maximises total effective satisfaction while staying within the distance budget. Apply satisfaction decay ($k = 0.1$) based on cumulative distance before each visit, and apply a 10% penalty to any location visited beyond the category threshold.

Dry Run — Route: Start → Museum → Park → End

Distance accumulation:

- Start → Museum: 3 km. Cumulative before Museum = 0 km.
- Museum → Park: 4 km. Cumulative before Park = 3 km.
- Park → End: 5 km.
- Total distance = $3 + 4 + 5 = 12 \text{ km} \leq 20 \text{ km}$. ✓

Effective scores (no penalties; each category visited only once):

$$S_{\text{Museum}} = 8 \times e^{-0.1 \times 0} = 8.000$$

$$S_{\text{Park}} = 6 \times e^{-0.1 \times 3} = 6 \times 0.7408 = 4.445$$

$$\text{Total Score} = 8.000 + 4.445 = \mathbf{12.445}$$

Answer Key

Optimal Route: Start $\rightarrow$ Museum $\rightarrow$ Park $\rightarrow$ End **Total Distance:** 12 km **Total Score:** 12.445

1.2 Public Test Case 2 *Easy*

Parameters

Start	Home
End	Office
Distance Budget	15 km
Category Threshold (n)	1

Locations

#	Name	Score	Category
0	Home	—	START
1	Cafe	7	Food
2	Waterfall	9	Nature
3	Office	—	END

Distance Matrix (km)

	Home	Cafe	Waterfall	Office
Home	0	2	6	12
Cafe	2	0	5	10
Waterfall	6	5	0	6
Office	12	10	6	0

Task. Same as above.

Dry Run — Route: Home $\rightarrow$ Cafe $\rightarrow$ Waterfall $\rightarrow$ Office

Distance accumulation:

- Home $\rightarrow$ Cafe: 2 km. Cumulative before Cafe = 0 km.
- Cafe $\rightarrow$ Waterfall: 5 km. Cumulative before Waterfall = 2 km.
- Waterfall $\rightarrow$ Office: 6 km.
- Total distance = $2 + 5 + 6 = 13 \text{ km} \leq 15 \text{ km}$. ✓

Effective scores (no penalties; Food and Nature are distinct categories):

$$S_{\text{Cafe}} = 7 \times e^{-0.1 \times 0} = 7.000$$

$$S_{\text{Waterfall}} = 9 \times e^{-0.1 \times 2} = 9 \times 0.8187 = 7.369$$

$$\text{Total Score} = 7.000 + 7.369 = 14.369$$

Answer Key

Optimal Route: Home $\rightarrow$ Cafe $\rightarrow$ Waterfall $\rightarrow$ Office **Total Distance:** 13 km **Total Score:** 14.369

1.3 Public Test Case 3 *Medium***Parameters**

Start	City Center
End	Beach
Distance Budget	25 km
Category Threshold (n)	1

Locations

#	Name	Score	Category
0	City Center	—	START
1	Fort	10	Historical
2	Temple	8	Historical
3	Garden	7	Nature
4	Beach	—	END

Distance Matrix (km)

	City C.	Fort	Temple	Garden	Beach
City Center	0	4	6	8	15
Fort	4	0	3	6	12
Temple	6	3	0	4	10
Garden	8	6	4	0	7
Beach	15	12	10	7	0

Task. Find the route that maximises total effective satisfaction. Note: category threshold is 1, so visiting two Historical locations will incur a penalty on the second.

Dry Run — Route: City Center $\rightarrow$ Fort $\rightarrow$ Temple $\rightarrow$ Garden $\rightarrow$ Beach**Distance accumulation:**

- City Center $\rightarrow$ Fort: 4 km. Cumulative before Fort = 0 km.
- Fort $\rightarrow$ Temple: 3 km. Cumulative before Temple = 4 km.
- Temple $\rightarrow$ Garden: 4 km. Cumulative before Garden = 7 km.
- Garden $\rightarrow$ Beach: 7 km.
- Total distance = $4 + 3 + 4 + 7 = 18 \text{ km} \leq 25 \text{ km}$. ✓

Category tracking ($n = 1$): Fort is the 1st Historical location (no penalty). Temple is the 2nd Historical location $\rightarrow$ 10% penalty applied.

Effective scores:

$$S_{\text{Fort}} = 10 \times e^{-0.1 \times 0} = 10.000$$

$$S_{\text{Temple}} = 8 \times e^{-0.1 \times 4} \times 0.9 = 8 \times 0.6703 \times 0.9 = 4.826$$

$$S_{\text{Garden}} = 7 \times e^{-0.1 \times 7} = 7 \times 0.4966 = 3.476$$

$$\text{Total Score} = 10.000 + 4.826 + 3.476 = \mathbf{18.302}$$

Answer Key

Optimal Route: City Center → Fort → Temple → Garden → Beach **Total Distance:** 18 km
Total Score: 18.302

1.4 Public Test Case 4 *Medium*

Parameters

Start	Station
End	Airport
Distance Budget	30 km
Category Threshold (n)	1

Locations

#	Name	Score	Category
0	Station	—	START
1	Zoo	9	Nature
2	Restaurant	6	Food
3	Mosque	8	Historical
4	Mall	5	Food
5	Airport	—	END

Distance Matrix (km)

	Station	Zoo	Rest.	Mosque	Mall	Airport
Station	0	5	3	7	4	18
Zoo	5	0	6	5	8	14
Restaurant	3	6	0	4	2	15
Mosque	7	5	4	0	6	12
Mall	4	8	2	6	0	14
Airport	18	14	15	12	14	0

Task. Find the optimal route. Two Food locations are present; visiting both will incur a penalty on the second.

Dry Run — Route: Station → Restaurant → Mall → Mosque → Zoo → Airport

Distance accumulation:

- Station → Restaurant: 3 km. Cumulative before Restaurant = 0 km.
- Restaurant → Mall: 2 km. Cumulative before Mall = 3 km.
- Mall → Mosque: 6 km. Cumulative before Mosque = 5 km.
- Mosque → Zoo: 5 km. Cumulative before Zoo = 11 km.
- Zoo → Airport: 14 km.
- Total distance = 3 + 2 + 6 + 5 + 14 = 30 km ≤ 30 km. ✓

Category tracking ($n = 1$): Restaurant is the 1st Food (no penalty). Mall is the 2nd Food $\rightarrow$ 10% penalty. Mosque (1st Historical) and Zoo (1st Nature): no penalty.

Effective scores:

$$S_{\text{Restaurant}} = 6 \times e^{-0.1 \times 0} = 6.000$$

$$S_{\text{Mall}} = 5 \times e^{-0.1 \times 3} \times 0.9 = 5 \times 0.7408 \times 0.9 = 3.334$$

$$S_{\text{Mosque}} = 8 \times e^{-0.1 \times 5} = 8 \times 0.6065 = 4.852$$

$$S_{\text{Zoo}} = 9 \times e^{-0.1 \times 11} = 9 \times 0.3329 = 2.996$$

$$\text{Total Score} = 6.000 + 3.334 + 4.852 + 2.996 = \mathbf{17.182}$$

Answer Key

Optimal Route: Station $\rightarrow$ Restaurant $\rightarrow$ Mall $\rightarrow$ Mosque $\rightarrow$ Zoo $\rightarrow$ Airport **Total Distance:** 30 km **Total Score:** 17.182

1.5 Public Test Case 5 *Hard*

Parameters

Start	Hotel
End	Hilltop
Distance Budget	35 km
Category Threshold (n)	1

Locations

#	Name	Score	Category
0	Hotel	—	START
1	Castle	10	Historical
2	Ruins	8	Historical
3	Lake	9	Nature
4	Forest	7	Nature
5	Bistro	6	Food
6	Hilltop	—	END

Distance Matrix (km)

	Hotel	Castle	Ruins	Lake	Forest	Bistro	Hilltop
Hotel	0	5	8	6	10	3	22
Castle	5	0	4	7	9	6	18
Ruins	8	4	0	5	7	8	16
Lake	6	7	5	0	4	7	14
Forest	10	9	7	4	0	9	12
Bistro	3	6	8	7	9	0	20
Hilltop	22	18	16	14	12	20	0

Task. Multiple categories with threshold 1. Decay and penalty interactions are significant. Find the route that maximises effective satisfaction within the budget.

Dry Run — Route: Hotel → Bistro → Castle → Ruins → Lake → Forest → Hilltop**Distance accumulation:**

- Hotel → Bistro: 3 km. Cumulative before Bistro = 0 km.
- Bistro → Castle: 6 km. Cumulative before Castle = 3 km.
- Castle → Ruins: 4 km. Cumulative before Ruins = 9 km.
- Ruins → Lake: 5 km. Cumulative before Lake = 13 km.
- Lake → Forest: 4 km. Cumulative before Forest = 18 km.
- Forest → Hilltop: 12 km.
- Total distance = $3 + 6 + 4 + 5 + 4 + 12 = 34 \text{ km} \leq 35 \text{ km}$. ✓

Category tracking ($n = 1$): Bistro (1st Food), Castle (1st Historical), Ruins (2nd Historical → penalty), Lake (1st Nature), Forest (2nd Nature → penalty).

Effective scores:

$$S_{\text{Bistro}} = 6 \times e^{-0.1 \times 0} = 6.000$$

$$S_{\text{Castle}} = 10 \times e^{-0.1 \times 3} = 10 \times 0.7408 = 7.408$$

$$S_{\text{Ruins}} = 8 \times e^{-0.1 \times 9} \times 0.9 = 8 \times 0.4066 \times 0.9 = 2.927$$

$$S_{\text{Lake}} = 9 \times e^{-0.1 \times 13} = 9 \times 0.2725 = 2.453$$

$$S_{\text{Forest}} = 7 \times e^{-0.1 \times 18} \times 0.9 = 7 \times 0.1653 \times 0.9 = 1.041$$

$$\text{Total Score} = 6.000 + 7.408 + 2.927 + 2.453 + 1.041 = \mathbf{19.83}$$

Answer Key

Optimal Route: Hotel → Bistro → Castle → Ruins → Lake → Forest → Hilltop **Total Distance:** 34 km **Total Score:** 19.83

1.6 Public Test Case 6 *Hard***Parameters**

Start	Depot
End	Lighthouse
Distance Budget	40 km
Category Threshold (n)	1

Locations

#	Name	Score	Category
0	Depot	—	START
1	Shrine	9	Historical
2	Monastery	8	Historical
3	Vineyard	7	Nature
4	Cliff	9	Nature
5	Bakery	5	Food
6	Diner	6	Food
7	Lighthouse	—	END

Distance Matrix (km)

	Depot	Shrine	Mon.	Vyd.	Cliff	Bakery	Diner	L.house
Depot	0	6	9	7	11	4	5	25
Shrine	6	0	4	8	10	7	8	20
Monastery	9	4	0	6	8	9	10	18
Vineyard	7	8	6	0	5	8	9	16
Cliff	11	10	8	5	0	10	11	14
Bakery	4	7	9	8	10	0	3	22
Diner	5	8	10	9	11	3	0	21
Lighthouse	25	20	18	16	14	22	21	0

Task. Largest Part A public test case with 6 intermediate locations across 3 categories. Balance decay, penalties, and budget carefully.

Dry Run — Route: Depot → Shrine → Monastery → Cliff → Vineyard → Lighthouse

Distance accumulation:

- Depot → Shrine: 6 km. Cumulative before Shrine = 0 km.
- Shrine → Monastery: 4 km. Cumulative before Monastery = 6 km.
- Monastery → Cliff: 8 km. Cumulative before Cliff = 10 km.
- Cliff → Vineyard: 5 km. Cumulative before Vineyard = 18 km.
- Vineyard → Lighthouse: 16 km.
- Total distance = $6 + 4 + 8 + 5 + 16 = 39 \text{ km} \leq 40 \text{ km}$. ✓

Category tracking ($n = 1$): Shrine (1st Historical), Monastery (2nd Historical → penalty), Cliff (1st Nature), Vineyard (2nd Nature → penalty). Bakery and Diner are skipped to stay within budget and maximise score.

Effective scores:

$$S_{\text{Shrine}} = 9 \times e^{-0.1 \times 0} = 9.000$$

$$S_{\text{Monastery}} = 8 \times e^{-0.1 \times 6} \times 0.9 = 8 \times 0.5488 \times 0.9 = 3.951$$

$$S_{\text{Cliff}} = 9 \times e^{-0.1 \times 10} = 9 \times 0.3679 = 3.311$$

$$S_{\text{Vineyard}} = 7 \times e^{-0.1 \times 18} \times 0.9 = 7 \times 0.1653 \times 0.9 = 1.041$$

$$\text{Total Score} = 9.000 + 3.951 + 3.311 + 1.041 = \mathbf{17.303}$$

Answer Key

Optimal Route: Depot → Shrine → Monastery → Cliff → Vineyard → Lighthouse **Total Distance:** 39 km **Total Score:** 17.303

Part B: Dynamic Ride Sharing with Flexible Routing

Background. A ride-sharing system dynamically computes and updates routes as new ride requests arrive. Each request has a pickup and drop-off location and a flexibility constraint.

Flexibility Constraint: For each request i with pickup P_i and dropoff D_i :

$$d(P_i \rightarrow D_i \text{ via route}) \leq \text{base_dist}_i + \Delta_i$$

Pickup Order: Pickup must always appear before dropoff in the route.

Capacity: Vehicle occupancy must never exceed capacity C at any point.

Objective: Minimise total travel distance while satisfying all constraints.

2.1 Public Test Case 1 *Easy*

Parameters

Start (Source)	S
Destination	E
Vehicle Capacity C	2
Number of Requests	1

Requests

#	Pickup	Dropoff	Base Dist (km)	Flexibility Δ
1	P1	D1	5	2

Distance Matrix (km)

	S	P1	D1	E
S	0	4	11	15
P1	4	0	5	12
D1	11	5	0	5
E	15	12	5	0

Task. Compute the optimal route from S to E that services the ride request, minimises total travel distance, and satisfies the flexibility constraint. Report the final route sequence, total distance, and passenger count at each stop.

Dry Run — Route: S $\rightarrow$ P1 $\rightarrow$ D1 $\rightarrow$ E

Distance calculation:

$$S \rightarrow P1 = 4, \quad P1 \rightarrow D1 = 5, \quad D1 \rightarrow E = 5$$

$$\text{Total} = 4 + 5 + 5 = 14 \text{ km}$$

Flexibility check for Request 1:

$$d(P1 \rightarrow D1) = 5 \leq 5 + 2 = 7 \quad \checkmark$$

Passenger count:

Location	Event	Passengers
S	—	0
P1	Pickup R1	1
D1	Dropoff R1	0
E	—	0

Answer Key**Optimal Route:** $S \rightarrow P1 \rightarrow D1 \rightarrow E$ **Total Distance:** 14 km**2.2 Public Test Case 2** *Easy***Parameters**

Start	S
Destination	E
Vehicle Capacity C	3
Number of Requests	1

Requests

#	Pickup	Dropoff	Base Dist (km)	Flexibility Δ
1	P1	D1	7	0

Distance Matrix (km)

	S	P1	D1	E
S	0	3	9	14
P1	3	0	7	12
D1	9	7	0	6
E	14	12	6	0

Task. Zero flexibility ($\Delta = 0$) means the system must service this request on the direct path only — no intermediate stops between P1 and D1 are permissible.

Dry Run — Route: $S \rightarrow P1 \rightarrow D1 \rightarrow E$ **Distance calculation:**

$$S \rightarrow P1 = 3, \quad P1 \rightarrow D1 = 7, \quad D1 \rightarrow E = 6$$

$$\text{Total} = 3 + 7 + 6 = 16 \text{ km}$$

Flexibility check for Request 1:

$$d(P1 \rightarrow D1) = 7 \leq 7 + 0 = 7 \quad \checkmark \quad (\text{exactly at limit})$$

Since $\Delta = 0$, no detour is allowed whatsoever. The direct $P1 \rightarrow D1$ leg is taken, satisfying the constraint with zero slack.

Passenger count:

Location	Event	Passengers
S	—	0
P1	Pickup R1	1
D1	Dropoff R1	0
E	—	0

Answer Key**Optimal Route:** $S \rightarrow P1 \rightarrow D1 \rightarrow E$ **Total Distance:** 16 km**2.3 Public Test Case 3** *Easy***Parameters**

Start	S
Destination	E
Vehicle Capacity C	2
Number of Requests	2

Requests

#	Pickup	Dropoff	Base Dist (km)	Flexibility Δ
1	P1	D1	6	1
2	P2	D2	5	2

Distance Matrix (km)

	S	P1	D1	P2	D2	E
S	0	3	8	11	15	20
P1	3	0	6	9	13	17
D1	8	6	0	4	8	12
P2	11	9	4	0	5	9
D2	15	13	8	5	0	6
E	20	17	12	9	6	0

Task. Service both requests sequentially (no simultaneous overlap). Report route, total distance, and passenger count at each stop.

Dry Run — Route: $S \rightarrow P1 \rightarrow D1 \rightarrow P2 \rightarrow D2 \rightarrow E$ **Distance calculation:**

$$S \rightarrow P1 = 3, P1 \rightarrow D1 = 6, D1 \rightarrow P2 = 4, P2 \rightarrow D2 = 5, D2 \rightarrow E = 6$$

$$\text{Total} = 3 + 6 + 4 + 5 + 6 = 24 \text{ km}$$

Flexibility checks:

$$R1: d(P1 \rightarrow D1) = 6 \leq 6 + 1 = 7 \quad \checkmark$$

$$R2: d(P2 \rightarrow D2) = 5 \leq 5 + 2 = 7 \quad \checkmark$$

Requests are served sequentially so passengers never overlap; at no point does occupancy exceed $C = 2$.

Passenger count:

Location	Event	Passengers
S	—	0
P1	Pickup R1	1
D1	Dropoff R1	0
P2	Pickup R2	1
D2	Dropoff R2	0
E	—	0

Answer Key

Optimal Route: $S \rightarrow P1 \rightarrow D1 \rightarrow P2 \rightarrow D2 \rightarrow E$

Total Distance: 24 km

2.4 Public Test Case 4 *Easy*

Parameters

Start	S
Destination	E
Vehicle Capacity C	2
Number of Requests	2

Requests

#	Pickup	Dropoff	Base Dist (km)	Flexibility Δ
1	P1	D1	6	3
2	P2	D2	5	1

Distance Matrix (km)

	S	P1	D1	P2	D2	E
S	0	4	9	6	12	18
P1	4	0	6	5	9	14
D1	9	6	0	7	4	10
P2	6	5	7	0	5	13
D2	12	9	4	5	0	8
E	18	14	10	13	8	0

Task. A second request arrives dynamically. Determine the optimal route servicing both requests while satisfying all constraints. Check whether interleaving is feasible.

Dry Run — Route: $S \rightarrow P1 \rightarrow D1 \rightarrow P2 \rightarrow D2 \rightarrow E$

Interleaved attempt first ($S \rightarrow P1 \rightarrow P2 \rightarrow D1 \rightarrow D2 \rightarrow E$):

R2 path: $d(P2 \rightarrow D1 \rightarrow D2) = 7 + 4 = 11 > 5 + 1 = 6 \quad \times \text{ FAILS}$

All other interleaved orderings similarly violate R2's tight flexibility constraint ($\Delta = 1$). Hence sequential routing is the only feasible option.

Sequential route — distance calculation:

$$S \rightarrow P1 = 4, P1 \rightarrow D1 = 6, D1 \rightarrow P2 = 7, P2 \rightarrow D2 = 5, D2 \rightarrow E = 8$$

$$\text{Total} = 4 + 6 + 7 + 5 + 8 = 30 \text{ km}$$

Flexibility checks:

$$\text{R1: } d(P1 \rightarrow D1) = 6 \leq 6 + 3 = 9 \quad \checkmark$$

$$\text{R2: } d(P2 \rightarrow D2) = 5 \leq 5 + 1 = 6 \quad \checkmark$$

Passenger count:

Location	Event	Passengers
S	—	0
P1	Pickup R1	1
D1	Dropoff R1	0
P2	Pickup R2	1
D2	Dropoff R2	0
E	—	0

Answer Key

Optimal Route: $S \rightarrow P1 \rightarrow D1 \rightarrow P2 \rightarrow D2 \rightarrow E$

Total Distance: 30 km

2.5 Public Test Case 5 *Medium*

Parameters

Start	S
Destination	E
Vehicle Capacity C	2
Number of Requests	2

Requests

#	Pickup	Dropoff	Base Dist (km)	Flexibility Δ
1	A	B	7	2
2	C	D	7	3

Distance Matrix (km)

	S	A	C	B	D	E
S	0	4	7	10	14	22
A	4	0	4	7	11	18
C	7	4	0	4	7	15
B	10	7	4	0	5	12
D	14	11	7	5	0	8
E	22	18	15	12	8	0

Task. Two requests with overlapping regions are to be interleaved. Find the optimal interleaved route. Verify all constraints including simultaneous occupancy does not exceed $C = 2$.

Dry Run — Route: $S \rightarrow A \rightarrow C \rightarrow B \rightarrow D \rightarrow E$

Distance calculation:

$$S \rightarrow A = 4, A \rightarrow C = 4, C \rightarrow B = 4, B \rightarrow D = 5, D \rightarrow E = 8$$

$$\text{Total} = 4 + 4 + 4 + 5 + 8 = 25 \text{ km}$$

Flexibility checks:

$$\text{R1 (A} \rightarrow \text{B via C): } d(A \rightarrow C \rightarrow B) = 4 + 4 = 8 \leq 7 + 2 = 9 \quad \checkmark$$

$$\text{R2 (C} \rightarrow \text{D via B): } d(C \rightarrow B \rightarrow D) = 4 + 5 = 9 \leq 7 + 3 = 10 \quad \checkmark$$

Capacity check: After pickup at A, occupancy = 1. After pickup at C, occupancy = 2 $\leq C = 2$ $\checkmark$. After dropoff at B, occupancy = 1. After dropoff at D, occupancy = 0.

Passenger count:

Location	Event	Passengers
S	—	0
A	Pickup R1	1
C	Pickup R2	2
B	Dropoff R1	1
D	Dropoff R2	0
E	—	0

Answer Key

Optimal Route: $S \rightarrow A \rightarrow C \rightarrow B \rightarrow D \rightarrow E$

Total Distance: 25 km

2.6 Public Test Case 6 *Medium*

Parameters

Start	S
Destination	E
Vehicle Capacity C	1
Number of Requests	2

Requests

#	Pickup	Dropoff	Base Dist (km)	Flexibility Δ
1	P1	D1	7	2
2	P2	D2	4	2

Distance Matrix (km)

	S	P1	D1	P2	D2	E
S	0	4	10	8	14	20
P1	4	0	7	5	11	16
D1	10	7	0	3	5	10
P2	8	5	3	0	4	13
D2	14	11	5	4	0	7
E	20	16	10	13	7	0

Task. With $C = 1$, simultaneous passengers are forbidden. Determine the only feasible routing strategy and verify all constraints.

Dry Run — Route: $S \rightarrow P1 \rightarrow D1 \rightarrow P2 \rightarrow D2 \rightarrow E$

Why interleaving is impossible: $C = 1$ means only one passenger may be aboard at a time. Any interleaved ordering (e.g. $S \rightarrow P1 \rightarrow P2 \rightarrow \dots$) would require P2 to be picked up before R1 is dropped off, pushing occupancy to 2. This violates the capacity constraint for every possible interleaved permutation.

Sequential route — distance calculation:

$$S \rightarrow P1 = 4, P1 \rightarrow D1 = 7, D1 \rightarrow P2 = 3, P2 \rightarrow D2 = 4, D2 \rightarrow E = 7$$

$$\text{Total} = 4 + 7 + 3 + 4 + 7 = 25 \text{ km}$$

Flexibility checks:

$$R1: d(P1 \rightarrow D1) = 7 \leq 7 + 2 = 9 \quad \checkmark$$

$$R2: d(P2 \rightarrow D2) = 4 \leq 4 + 2 = 6 \quad \checkmark$$

Passenger count:

Location	Event	Passengers
S	—	0
P1	Pickup R1	1
D1	Dropoff R1	0
P2	Pickup R2	1
D2	Dropoff R2	0
E	—	0

Maximum occupancy at any point = $1 \leq C = 1$. $\checkmark$

Answer Key

Optimal Route: $S \rightarrow P1 \rightarrow D1 \rightarrow P2 \rightarrow D2 \rightarrow E$

Total Distance: 25 km